

Report on Standard Swing Form Definition and Automated Diagnostic Logic Based on 2D Side-View Video Analysis in Youth Baseball

To build an image analysis and automated scoring system for youth baseball batting forms, it is essential to establish an algorithm that converts skeletal keypoints (Pose Estimation) obtained from a single-lens 2D side-view camera into comprehensive kinematic metrics and quantitatively evaluates deviations from the ideal kinematic sequence¹. This report defines standard swing mechanics tailored to the anatomical and physiological characteristics of youth players (elementary school age), formulates mathematical calculation models from 2D side-view images, specifies automated detection rules for representative swing faults, and presents a structured automated feedback system tied to scoring and corrective drills.

Standard Mechanics and Phase-by-Phase Kinematics of Youth Baseball Swings

While the batting motion is a continuous action, dividing the movement into five sequential phases in 2D side-view image analysis allows for effective evaluation of relative body positioning (alignment) and joint angles at each phase³.

1. Setup / Stance

The primary objective during the preparation phase is to eliminate unnecessary muscular tension and establish a power position to store potential mechanical energy⁶. Stance width is set slightly wider than shoulder width, with weight distributed over the balls of both feet (first metatarsal heads)⁶. Knees are squeezed slightly inward rather than flaring outward, maintaining tension in the inner thighs to enhance lower-body rigidity⁹.

When flexing the hips slightly to create a forward trunk tilt, the ideal position is where the shin angle and the torso axis angle are parallel (Shin-Torso Parallelism), which activates the posterior thigh muscles (hamstrings) and gluteal muscle group⁷. The grip is positioned at ear to shoulder height, approximately 1 to 2 ball-widths outside the rear toe, preventing the bat from casting outward and ensuring an initial swing plane that is level to slightly downward⁷.

2. Loading / Stride Phase

This phase involves linear translational movement toward the pitcher while simultaneously storing rotational power (loading) over the rear leg³. As the front foot lifts, weight shifts onto the rear hip joint, engaging the pelvis⁶. It is critical that the rear knee does not sway outward past the outer edge of the foot (preventing excessive negative weight shift)³.

As the lead foot strides forward, the grip and upper body remain loaded over the rear leg, stretching the front scapula (Scapular Load)³. This opposing movement creates hip-shoulder separation (torsional difference)³. During the stride, horizontal movement of the head (line of sight) should be minimized, maintaining a smooth parallel translation of the body axis⁶.

3. Foot Strike / Foot Plant

The explosive rotational movement of the swing begins the moment the front heel plants (Heel Plant)³. As the lead heel touches down, flexion of the lead knee joint stops, forming a firm front-side wall (braking mechanism)³. This ground reaction force (exceeding approximately 123% of body weight) drives momentum transfer from the lower body to the upper body¹¹. A proper kinematic sequence is initiated: the pelvis (hip) reaches maximum rotational velocity first (approx. 714 deg/s), followed sequentially by the thorax/shoulders (approx. 937 deg/s), hands, and bat (approx. 31 m/s)¹¹. At the start of rotation, the angle between the lead forearm and torso (Early Connection) is maintained between 80° and 105°, placing the bat onto the rotational plane of the torso¹⁶.

4. Impact Phase

This phase focuses on transferring stored rotational energy to the ball with maximum efficiency¹³. At impact, seven critical checkpoints must be satisfied: top hand palm-up / bottom hand palm-down, rear foot on toe, firm lead leg wall, rear elbow slotted near the torso, head positioned inside the triangle between both knees, a straight structural line connecting the head, rear knee, and ground, and eyes locked on the contact point¹⁷.

To align the bat path with the incoming ball flight (slightly downward angle), an optimal attack angle (vertical tilt at contact) ranges from +5° to +15° (slight upward angle)¹⁶. An attack angle exceeding +20° reduces exit velocity and leads to pop-ups¹⁹.

5. Follow-through Phase

This is the deceleration phase where inertia is managed without forcing a sudden stop, stabilizing swing trajectory⁶. Rather than forcibly rolling the wrists early, the bat head should be driven extended toward center field¹³. Maintaining the palm-up / palm-down relationship through contact extends the hitting zone (flat spot)³. The forward trunk tilt and head position established at impact should be preserved, finishing in a balanced posture supported between both feet¹⁴.

Swing Phase	Target Joints / Body Segments	Baseline Parameter / Kinematic Target	Physiological / Biomechanical Function
Setup	Hips, Knees, Shin angle, Torso angle,	Shin-torso parallelism angle	Posterior chain activation, Prevention of

	Grip	difference $\leq$, Grip height at ear-to-shoulder level ²¹	casting ²¹
Stride	Rear knee, Rear hip, Head, Grip	Rear knee inside rear foot boundary, Minimal head displacement ³	Energy containment, Hip-shoulder separation loading ⁶
Foot Strike	Lead knee, Pelvis angle, Shoulder angle, Forearm-torso angle	Lead knee stabilization/extensi on, Early Connection 80° to 105° ³	Ground reaction force capture, Kinematic sequence initiation ³
Impact	Bat tilt angle, Both elbows, Lead leg, Head axis	Attack angle +5° to +15°, Rear elbow slotted in ¹⁶	Ball path matching, Peak energy transfer ¹⁶
Follow-through	Wrist joints, Center of gravity, Torso tilt	Elimination of early wrist roll, Preserved trunk tilt ¹³	Extension of hitting zone, Trajectory stabilization ¹³

Mathematical Models for Pose Estimation and Kinematic Metrics from 2D Side-View Images

Coordinates (x, y) obtained from single-lens 2D side-view video (e.g., smartphones) vary based on camera distance, body size, and anatomical individualities². Therefore, raw pixel coordinates must be transformed into scale-invariant angular metrics and normalized relative coordinate systems¹⁰.

1. Angle Calculation Logic for 2D Vectors

Vectors connecting keypoints are defined, and joint/segment angles are derived using dot products and the two-argument arctangent function ($\arctan2$)¹⁰. The inclination angle θ of a vector connecting two keypoints $A(x_a, y_a)$ and $B(x_b, y_b)$ relative to the horizontal axis is defined as:

$$\theta = \arctan2(y_b - y_a, x_b - x_a)$$

The joint angle ϕ formed by three points A, B, C (e.g., shoulder, elbow, wrist) is calculated via the dot product of vectors BA and BC :

$$\phi = \arccos \left(\frac{BA \cdot BC}{\|BA\| \|BC\|} \right)$$

2. Definition of Key Kinematic Metrics from Side View

- **Shin-Torso Parallelism Angle (θ_{ST})**: Absolute angle difference between the shin vector V_{shin} (ankle to knee) and torso vector V_{torso} (hip to shoulder)²¹.

$$\theta_{ST} = |\theta_{shin} - \theta_{torso}|$$

Maintaining $\theta_{ST} \leq 10^\circ$ from setup through stride indicates proper power position maintenance²¹.

- **Early Connection Angle (θ_{EC})**: Angle between the torso vector V_{torso} and lead forearm vector V_{lead_arm} (lead shoulder to lead wrist)¹¹. Evaluated at rotation initiation (first frame of trunk rotation), with an optimal range of $80^\circ \leq \theta_{EC} \leq 105^\circ$ ¹¹.
- **Lead Knee Blocking Index ($\Delta\phi_{knee}$)**: Change in lead knee angle between Foot Strike $\phi_{knee}(t_{fs})$ and Impact $\phi_{knee}(t_{imp})$ ⁶.

$$\Delta\phi_{knee} = \phi_{knee}(t_{imp}) - \phi_{knee}(t_{fs})$$

A value of $\Delta\phi_{knee} \geq 0$ (knee maintaining angle or extending/bracing) indicates proper blocking, whereas $\Delta\phi_{knee} < 0$ (knee flexing further) signifies collapse⁶.

- **Head Translation Ratio (R_{head})**: Horizontal displacement of the head X -coordinate

from setup (X_{head_setup}) to impact (X_{head_impact}), normalized by torso length (L_{torso} , distance between shoulder and hip)².

$$R_{head} = \frac{X_{head_impact} - X_{head_setup}}{L_{torso}}$$

A ratio $R_{head} >$ indicates excessive forward lunging (rushing)¹⁴.

- **Estimated Attack Angle (θ_{AA}):** Tangential angle of the quadratic curve fit to wrist/bat trajectory across key frames around impact¹⁶. Horizontal is 0° , upward tilt is positive, and downward tilt is negative²³.

Metric Name	Target Keypoint Group	Mathematical Calculation / Vector Definition	Baseline Range / Target Value
Shin-Torso Parallelism	Both ankles, knees, hips, shoulders	Angle difference θ_{ST} between shin and torso vectors	$\leq$ (Setup to Stride) ²¹
Early Connection Angle	Lead shoulder, lead wrist, rear hip	Angle θ_{EC} of lead forearm vector relative to torso axis	$80^\circ \sim$ (Rotation start) ¹¹
Lead Knee Blocking	Lead hip, lead knee, lead ankle	Difference in knee flexion angle $\Delta\phi_{knee}$ (Impact vs. Foot Strike)	$\Delta\phi_{knee} \geq$ (Bracing/extension) ⁶
Head Vector Displacement	Head (ear/nose), rear ankle, torso length	Normalized horizontal head translation ratio R_{head}	$\leq$ (Stable rotational axis) ²

Estimated Attack Angle	Both wrists (grip), Bat tip	Tangential slope angle of grip/barrel trajectory at impact	$+5^{\circ} \sim$ (Slight upper tilt) ¹⁶
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Common Swing Faults in Youth Baseball and Automated Detection Rules

Batting faults in elementary school players often stem from undeveloped physical strength, overly heavy equipment (bats), or misinterpretations of abstract verbal cues (e.g., "chop down on the ball," "don't open up")¹⁵. Rule-based detection algorithms for 2D side-view analysis are detailed below.

1. Door Swing / Casting

When a player swings without adequate forward trunk tilt (upright stance), centrifugal force pulls the grip and bat away from the body, producing an overly wide, sweeping arc⁹. This is prevalent when players rely purely on arm strength or use bats that are too heavy¹⁵. Detection

logic: Early Connection Angle at rotation start triggers $\theta_{EC} >$, or the horizontal distance between wrist X -coordinate and chest X -coordinate $D_{wrist-chest}$ normalized by torso length exceeds threshold ($D_{wrist-chest} / L_{torso} >$)¹¹.

2. Forward Axis Drift / Rushing

This occurs when the upper body lunges forward toward the pitcher prior to foot strike due to insufficient loading into the rear hip³. This breaks the kinematic sequence, preventing

lower-body rotational power from transferring upward⁶. Detection logic: Rear knee X -coordinate sways past the outer rear ankle boundary during stride (excessive negative weight shift), or head translation ratio at impact exceeds $R_{head} >$ ³.

3. Arms-Only Swing / One-Piece Swing

Occurs when the lower body (pelvis) and upper body (thorax) fail to create a time lag (hip-shoulder separation), rotating simultaneously as a single rigid unit³. This bypasses the stretch-shortening cycle of the core torso muscles, resulting in low bat speed³. Detection logic: Zero temporal phase lag between pelvic vector rotation and shoulder vector rotation, or absence of dynamic elbow flexion angle changes during swing initiation⁶.

4. Excessive Upper Swing / Early Extension

Caused by an overemphasis on lifting the ball or losing forward trunk angle prior to impact as the pelvis thrusts forward (Early Extension)¹⁰. The bat head drops excessively, causing an over-exaggerated upward attack angle⁷. Detection logic: Calculated attack angle exceeds

$\theta_{AA} > \quad$, accompanied by significant backward trunk tilt relative to setup posture⁶.

5. Collapsed Lead Side

Occurs when the lead knee flares outward (opens) or continues to flex deeply post-landing, failing to brace against forward momentum⁶. Rotational energy leaks forward, causing axis

instability⁶. Detection logic: Lead knee angle change at impact is $\Delta\phi_{knee} < \quad$ (knee flexes further post-landing), or lead knee X -coordinate drifts forward past lead ankle X -coordinate⁶.

Swing Fault Pattern	Kinematic / Physical Root Cause	2D Side-View Detection Rules (Mathematical Conditions)	Impact on Motion & Ball Flight
Door Swing / Casting	Upright torso tilt, centrifugal casting, mismatched bat weight ⁹	$\theta_{EC} > \quad$ OR $D_{wrist-chest} / L_{torso}$ [cite: 11]	Bat casting, late contact point, weak hit ⁶
Forward Axis Drift	Poor rear hip load, lack of front-side braking ⁶	Impact $R_{head} > \quad$ OR $X_{head} > \quad$ [cite: 16, 17]	Reduced contact consistency, vulnerable to off-speed pitches ¹⁶
Arms-Only / One-Piece	Lack of hip-shoulder separation, weak core stability ³	Pelvis-shoulder rotation lag $\approx \quad$, Static elbow angle ⁶	Low bat speed, loss of exit velocity ⁶
Excessive Upper Swing	Loss of posture (Early Extension), over-lifting tendency ²⁰	$\theta_{AA} > \quad$ AND backward trunk collapse ¹⁸	Excessive pop-ups, scraping under the ball ⁷
Collapsed Lead Side	Insufficient front-leg strength, weak ground reaction braking ⁶	$\Delta\phi_{knee} < \quad$ OR lead knee forward drift ⁶	Rotational speed decay, contact inconsistency ⁶

Scoring Algorithm and Automated Pipeline Design

Overall swing score calculation (100-point scale) utilizes a phase-weighted deduction approach based on biomechanical impact¹¹. The processing pipeline begins by extracting 2D skeletal keypoints from video frames using pose estimation models (e.g., YOLOv8-pose, MediaPipe, or HRNet)¹. Time-series keypoint trajectories are then aligned to standard swing phases (Setup, Stride, Foot Strike, Impact, Follow-through) via Dynamic Time Warping (DTW) or key-pose detection models⁵. Once aligned, geometric vector angles and spatial parameters are computed and passed to the scoring engine and feedback module².

1. Phase-Based Weighting and Penalty Functions

Overall score S_{total} is defined as:

$$S_{total} = 100 - \sum_{k=1}^N W_k \cdot P_k(M_k, M_{ref,k})$$

Where W_k represents the weighting coefficient for evaluation metric k ($\sum W_k = 100$), M_k is the measured value, $M_{ref,k}$ is the target reference range, and P_k is a normalized penalty function (0.0 to 1.0) scaled to deviation magnitude.

Phase weighting breakdown based on dynamic contribution:

- **Setup Phase: Weight 10%** (Metrics: Shin-torso parallelism θ_{ST} , initial grip position)²¹
- **Stride Phase: Weight 20%** (Metrics: Rear knee sway, head movement stability)⁶
- **Foot Strike Phase: Weight 25%** (Metrics: Early Connection Angle θ_{EC} , hip-shoulder separation timing)⁶
- **Impact Phase: Weight 35%** (Metrics: Attack Angle θ_{AA} , lead knee blocking $\Delta\phi_{knee}$, head impact position)³
- **Follow-through Phase: Weight 10%** (Metrics: Torso tilt maintenance, finish balance)³

2. Feedback Generation Logic

The system identifies the metric with the largest score deduction (primary error) and outputs a three-tiered diagnostic report¹⁶:

1. **Calculated Scores:** Overall score and phase sub-scores.
2. **Point-by-Point Fault Description:** Specific anatomical misalignment and biomechanical cause⁷.
3. **Corrective Advice & Drills:** Tailored drills and actionable coaching cues⁹.

Fault-Specific Coaching Protocols and Corrective

Drills

To correct detected deviations, abstract instructions ("don't open up", "swing hard") are replaced with targeted drills and clear sensory cues¹⁶.

1. Corrective Protocol for Door Swing

Forcing players to "keep elbows tight" restricts wrist flexibility and causes rigidity²⁴. Instead, establishing proper forward posture enables the arms to naturally path inside-out relative to body rotation⁹.

- **Cross-Arm Rotation Drill:** Player crosses arms over the chest holding the bat against the torso, assumes proper forward posture, and rotates through the swing plane¹⁸. Teaches the relationship between torso tilt and rotational axis, preventing casting¹⁸.
- **Inside-Out Station Tee Drill:** Set tee on the inside corner low, prompting the player to hit the inner ball toward the opposite field (right field for righties)²⁸. Teaches leading with the grip to keep the path tight¹².

2. Corrective Protocol for Forward Axis Drift

Focuses on loading weight into the rear hip crease (groin area) to lock the pelvis, holding upper-body rotation until lead foot strike⁹.

- **5-Second Rear Hip Load Hold Drill:** Lift the lead leg and balance on the rear leg for 5 seconds before swinging¹². Builds core balance and rear-hip engagement to eliminate forward drift¹⁷.
- **Single-Leg Balance Swing Drill:** Swing while remaining balanced on the rear leg to constrain forward axial displacement¹⁷.

3. Corrective Protocol for Arms-Only / One-Piece Swing

Develops the feeling of sequential movement, where lower-body rotation leads and upper-body follows with a distinct delay¹³.

- **Chest-Hugged Bat Swing Drill:** Hug the bat across the chest so arms cannot operate independently, then rotate lower body to strike a ball off a tee²⁹. Directly connects lower-body force to the torso²².

4. Corrective Protocol for Excessive Upper Swing

Prevents dropping the rear shoulder and losing forward posture (Early Extension)¹⁹. Keeps the grip high to establish a level-to-slight-upper plane¹³.

- **High-Grip Freeze Drill:** Wide static stance without striding; swing and freeze at contact³⁰. Ensure the grip remains higher than the bat barrel upon impact, training barrel control and ideal attack angle.

Diagnosed Error	Automatically Generated	Kinematic Root Cause	Recommended Drill	Actionable Coaching Cue
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	Problem Description			
Door Swing	Bat casts away from the body, sweeping wide in a circle ⁶	Upright posture causing centrifugal casting, lack of inside-out path ¹⁸	Cross-Arm Rotation Drill ¹⁸ , Inside-Out Tee Drill ¹⁹	"Bow slightly forward and turn around your spine axis" ¹⁸
Forward Axis Drift	Body lunges forward, causing unstable contact timing ¹⁶	Lack of rear-hip loading, weak front-side brace ³	5-Sec Rear Hip Load Drill ¹⁷ , Single-Leg Swing ¹⁷	"Load into your back hip crease and count to 5" ¹⁷
Arms-Only Swing	Lower and upper body rotate together without lag ⁶	Poor hip-shoulder separation, arm-dominant motion ³	Chest-Hugged Bat Swing ²² , Tee Placement Drill ⁸	"Fire your hips first, let your arms follow behind" ³
Excessive Upper Swing	Undercutting the ball, producing excessive pop-ups ⁷	Early extension, rear shoulder dip, steep attack angle ⁷	High-Grip Freeze Drill, Hula-Hoop Drill ²¹	"Keep your hands higher than the bat head through impact"
Collapsed Lead Side	Lead knee flexes/collapses at impact, losing leverage ⁶	Weak lead-leg braking, failure to capture ground reaction force ³	Firm Lead-Leg Stop Drill, Single-Leg Swing ¹⁷	"Push the ground away with your front foot at contact" ³

Integrating these kinematic parameters, mathematical detection logic, and drill protocols into your analysis pipeline will enable your application to deliver precise, automated swing evaluation and actionable guidance for youth baseball players from standard 2D side-view video.

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